Supplementary Materials

GRRAS checklist and AI assisted technical verification

20 October 2026

Manuscript: *What can be measured from the broadcast? Validity of single-camera video for football-kick kinematics and swing-leg joint loads.* Author: Roberto Martín Gutiérrez, ORCID 0009-0004-5724-3502.
Revised after SportRxiv submission 1123.

Scope and provenance

The audit used the supplied 27-page preprint, the two-page GRRAS PDF, the three supplied Markdown checklists, the local manuscript sources and saved numerical outputs. The original submitted files were preserved. Repository base: commit 05d6ee8; original research release v1.0.0: DOI 10.5281/zenodo.23439408. The original archive is not described as containing these later corrections.

GRRAS reporting checklist

Item	Reporting issue	Location and clarification
1	Identify reliability or agreement in title or abstract	Abstract identifies a method-comparison study and reports agreement against optical motion capture.
2	Name the measurement method	Sections 2.2–2.4: RTMW-x, MotionAGFormer-B, virtual and real cameras, scaling and alignment.
3	Specify the target subject population	Introduction: football kicking; Sections 2.1 and 4.5 distinguish laboratory air kicks, two CMU ball-kick participants and one recreational field participant.
4	Specify the rater population	Automated computational method compared with motion capture; there are no sampled human raters. This does not imply reference-free or blinded evaluation.
5	State prior agreement knowledge and rationale	Introduction and Sections 4.1–4.3, with the relevant primary literature.
6	Explain sample size and replicates	Sections 2.1, 2.3 and 2.7: available-data sample; 83 MoVi kicks, seven CMU trials from two people, 200 3DSP shots; five views, five sizes, two frame rates; one MoVi noise seed and three CMU noise seeds (one for no-noise conditions).
7	Describe sampling	Section 2.1: 90 MoVi files, four without a usable kick, three further long-gap exclusions; 83 retained, eight with interpolated gaps up to 0.25 s. All seven listed CMU trials and all 200 3DSP shots.
8	Describe measurement process	Sections 2.2–2.6: kinematic windows, simplified inverse dynamics, detector residuals, camera, lifting, real video, twin, remedies and field example.
9	State measurement independence	Section 2.3 explicitly discloses reference-derived scale, sequence-wide rigid alignment, reference event timing and side in the known-event condition. The adapted lifter has participant-disjoint evaluation; this is not independent deployment calibration.
10	Describe statistics	Section 2.7: MAE, bias, descriptive LoA, ICC(A,1), clustered CIs, calibration slope, TOST and custom sign-flip waveform tests. Section 2.8 is a hypothetical event-count sensitivity only.
11	State actual numbers and replicates	83 MoVi participants; 249 typical participant-view pairs; 162 usable video recordings from 82 participants (80 with two cameras, two with one); seven CMU kicks from two people; field example: 13 recordings of one person.

Item	Reporting issue	Location and clarification
12	Describe sample characteristics	Sections 2.1 and 2.6: 55 female MoVi participants, non-football laboratory air kicks; CMU ball kicks; author field example with stated anthropometrics and views.
13	Report agreement and uncertainty	Tables 2–4 and aggregate outputs; Table 2 has participant-cluster CIs. The KLI clustered CI is added in Section 3.5 and this audit. Repeated sessions were not recorded, so individual longitudinal repeatability is not established.
14	Discuss practical relevance	Sections 4.1–4.6 retain the laboratory-to-football transfer limits and do not endorse clinical, scouting or injury-risk use.
15	Provide detailed results	Original public release: DOI 10.5281/zenodo.23439408 and GitHub. This PDF documents the subsequent technical checks and their limits. The revised supplementary document is this PDF, not a data archive.

Technical execution and coverage

The test suite completed with **63 passed, four dependency warnings**. Saved numerical outputs were re-aggregated on CPU; selected MAE, bias and absolute-agreement ICC values were independently recalculated with an ANOVA formula. Original GPU detector inference, full synthetic simulations and model training were not rerun. Passing tests does not establish clinical validity or prove author engagement.

The source-data flow was checked by opening all 90 available MoVi F-round files: 4 lacked a usable kicking action, 3 further trials had excessive gaps, 83 remained and 8 retained trials required short-gap interpolation. Participant-level files are not included in the Supplementary Materials PDF.

For KLI in the typical simulated condition, a new 1,000-resample participant-cluster bootstrap (seed 20261020; 83 participants, 249 pairs) gives ICC 95% CI 0.018 to 0.181. This is uncertainty for pooled method agreement, not injury-risk reliability.

Check	Claim or analysis	Audit outcome and limitation
1	Broadcast error range	Recomputed from saved detector measurements / bins; reported 7.6–12.9% (upper unrounded value 12.948%).
2	Noise model	Saved fit: sigma_frac 0.0631 and floor 3.886 px; rounded model is consistent.
3	Laboratory detector error	Saved laboratory summary gives 3.1–4.0% over the stated downscaled sizes; no new detector inference was run.
4	Joint error proportions	Saved detector summaries checked; ankle/wrist errors exceed hip/shoulder errors. This is comparison with annotations, not annotation-free truth.
5	Ball-contact offset	Five usable extrapolated-contact trials; saved mean 19.6 ms before contact, range 6.8–32.2 ms; feature change up to 2.1 degrees. This does not validate connected-ball sensor timing.
6	Noise-free and intensity analysis	Saved summaries and source filtering inspected; intensity correlation is descriptive and does not establish causality.
7	View-specific MAE	Recomputed: lateral 11.948, 135 degrees 13.462, 45 degrees 19.006, rear 20.038, frontal 22.650 degrees.
8	Height-specific MAE	Recomputed across the specified subset: 16.802–18.191 degrees over 50–400 px; aggregation is across the tested views.
9	Gaussian residual contrast	Saved ablation summaries checked; no new GPU simulations were executed.

Check	Claim or analysis	Audit outcome and limitation
10	Frame rate and event contrast	Saved ablation definitions and summaries checked; claims now describe the tested contrast without asserting that frame rate never matters.
11	Typical agreement table	18 independently recomputed MAE, bias and ICC values agree with the rounded table within 0.00011; 249 pairs are clustered within 83 participants.
12	Real-video table	24 feature-by-size cells recomputed from saved per-recording estimates; 162 recordings from 82 people, not 164 recordings. ICC 0.71 is moderate.
13	Digital twin	Recomputed across 48 aggregate cells: r 0.83838; mean simulated 12.9959 degrees, real 14.8459 degrees, mean absolute discrepancy 3.9145 degrees. Cells are not independent validation participants.
14	KLI association	Recomputed from reference moment z-scores: Spearman rho 0.794488; p 3.146e-19. Shared kinematic inputs preclude independent construct validation.
15	Calibration versus reliability	The saved KLI slope is 0.08099, with r 0.2043 and ICC 0.0903. The slope is scale-dependent and must not be labelled measured reliability.
16	48 / 590 injury counts	Arithmetic of the former formula reproduces approximately 48 / 590 when slope is substituted. That substitution is not justified. The 590 prediction is withdrawn; only independently assumed information fractions are illustrated.
17	League-season conversion	The 12–24 season prediction depended on the preceding unsupported step and assumed incidence; it is withdrawn.
18	Example model moments	Saved CMU output agrees with 1.008 vs 0.428 hip-flexor and 0.893 vs 0.118 knee-flexor N m/kg; simplified same-model estimates, not force-platform validation.
19	Replay fusion	Aggregate Table 7 checked: the reported single-view and fused estimates are retained as simulated laboratory results, without claiming demonstrated football scouting utility.
20	Ranking criterion	Table 9 retains knee ICC 0.80 with CI about 0.70–0.87. The criterion is post hoc and operational; it is not a clinically validated threshold.
21	Adapted lifter	Aggregate means recomputed: virtual MoVi 14.6769 to 8.2957 degrees; real 80 px 14.9565 to 13.3421; CMU 11.8216 to 16.7136. Evaluation uses saved predictions; training was not rerun.
22	Waveform inference	Table 10 and custom sign-flip implementation inspected. Correction is timewise within each of four waveform tests, not across all four; no claim of spm1d implementation is made.
23	Field means	Recomputed native lateral / rear-diagonal means: strong 11.9768 m/s, knee 93.0368 degrees, hip range 113.2271; soft 8.6803, 72.0823, 84.9809.
24	Field labels and contact validity	Strong/soft labels are author-specified. Ankle-speed contact remains a proxy; without an instrumented reference, true contact accuracy and force intensity cannot be certified. No fabricated human verification is recorded.

Bibliographic verification

The source BibTeX library has 52 entries; 46 are cited by the original manuscript. Metadata were checked using Crossref, PubMed/Europe PMC, arXiv and the primary websites. Crossref-only short-title discrepancies for Shan and Lakens were resolved against indexed or publisher full titles. Rate-limited Crossref records for Nunome, Dorge and Mendiguchia were checked against the DOI-indexed PubMed records. Identifiers and basic bibliographic identity were checked; this is not a systematic review or a full-text verification of every numerical claim in every cited work.

Abstracts and relevant accessible primary pages were examined where available. In particular, the revision corrects the interpretation of Koo and Li (ICC 0.71 is moderate), distinguishes gait-derived 5-degree error motivation from a football MAE standard, labels the 0.2-SD monitoring criterion as an exploratory screen, and removes the unjustified substitution of calibration slope into a reliability-based event formula. The Winter book metadata were verified; the exact 0.245 ratio is retained as a stated modelling assumption rather than claimed as independently checked against the book table. The official LaLiga announcement supports contact-timing availability but does not validate synchronisation of this pipeline.

Bibliographic checks during revision are summarized in this PDF; no machine-readable reference ledger is attached. Abstract review is not equated with full-text examination, and metadata verification is not presented as proof of novelty.

Revisions responding to moderation

Earlier disclosures did not clearly distinguish the author's work from AI assistance. The author developed the analysis code, generated scientific figures from the analyses, and authored the scientific content. Claude Code assisted with programming suggestions, debugging, English translation, and aesthetic refinements to the figures. The author reviewed these changes and takes responsibility for the research.

Additional scientific corrections include the real-video sample count; reference-assisted scale/alignment/event disclosures; the exploratory nature of operational thresholds and waveform inference; the model-derived nature of loads; the non-held-out posture-surrogate transfer evaluation; and withdrawal of the 590-injury and league-season predictions. No new injury outcomes or clinical validation are claimed.

Audit files and reproducibility

This Supplementary Materials PDF contains the GRRAS checklist and a description of the technical and bibliographic checks performed during revision. It does not include the underlying audit scripts, numeric ledgers, participant-level outputs or restricted model weights. The original research code, figures and aggregate results are available through the public GitHub repository and Zenodo release. This PDF is the supplementary document accompanying the revised manuscript.